Cloud Computing

Exercise-10

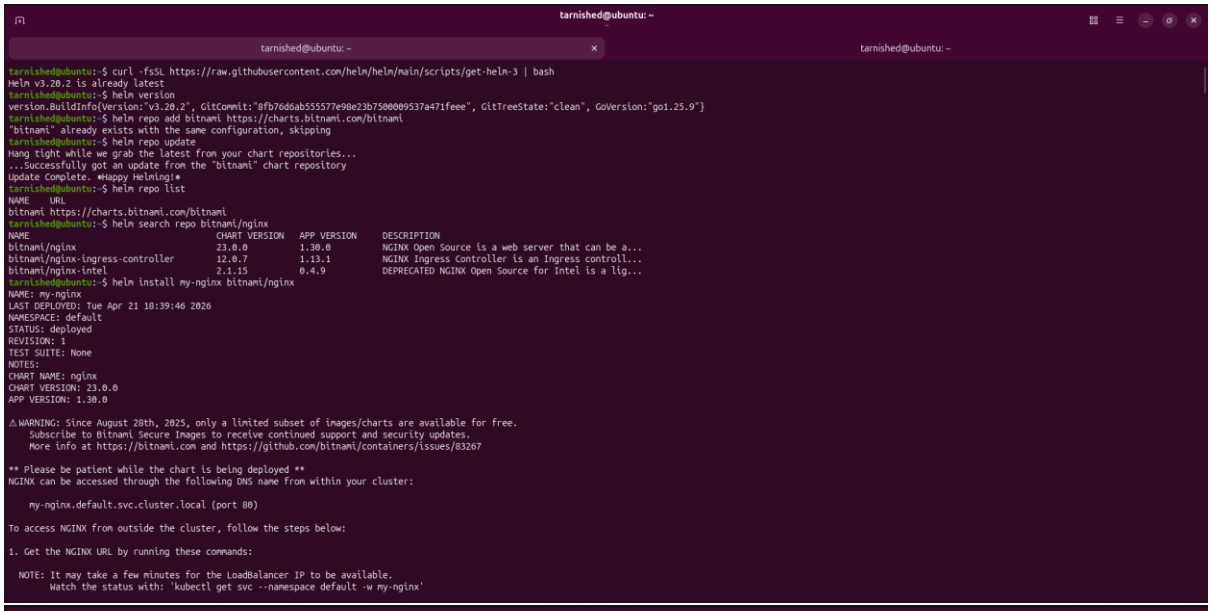
Name: CH. Aravind Kumar

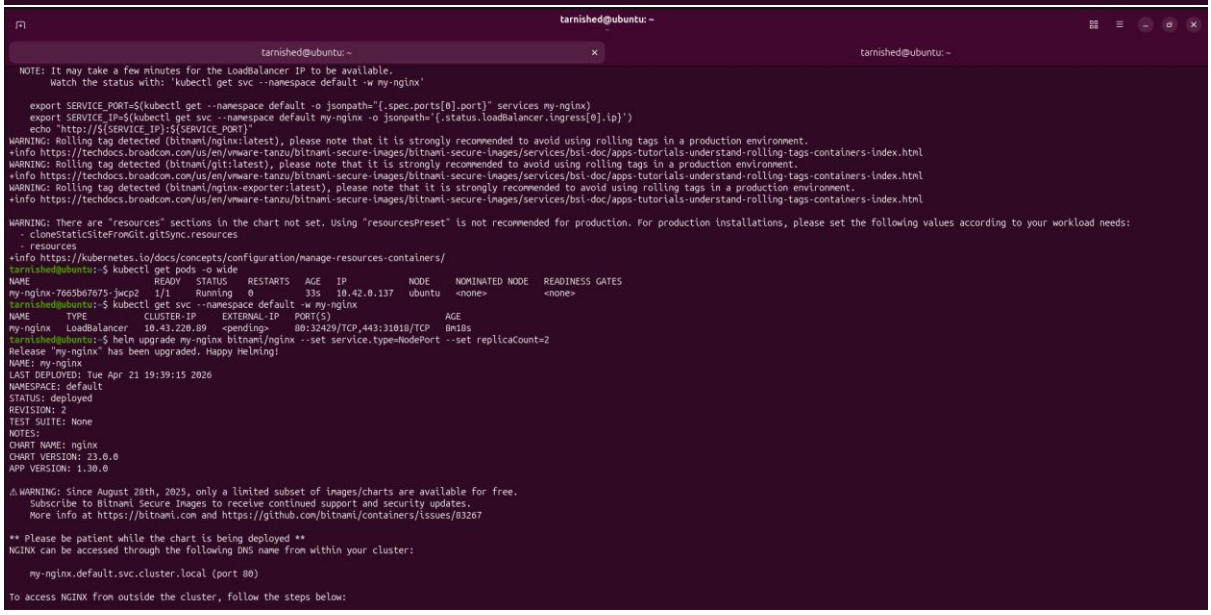
ID: 23MCCE11

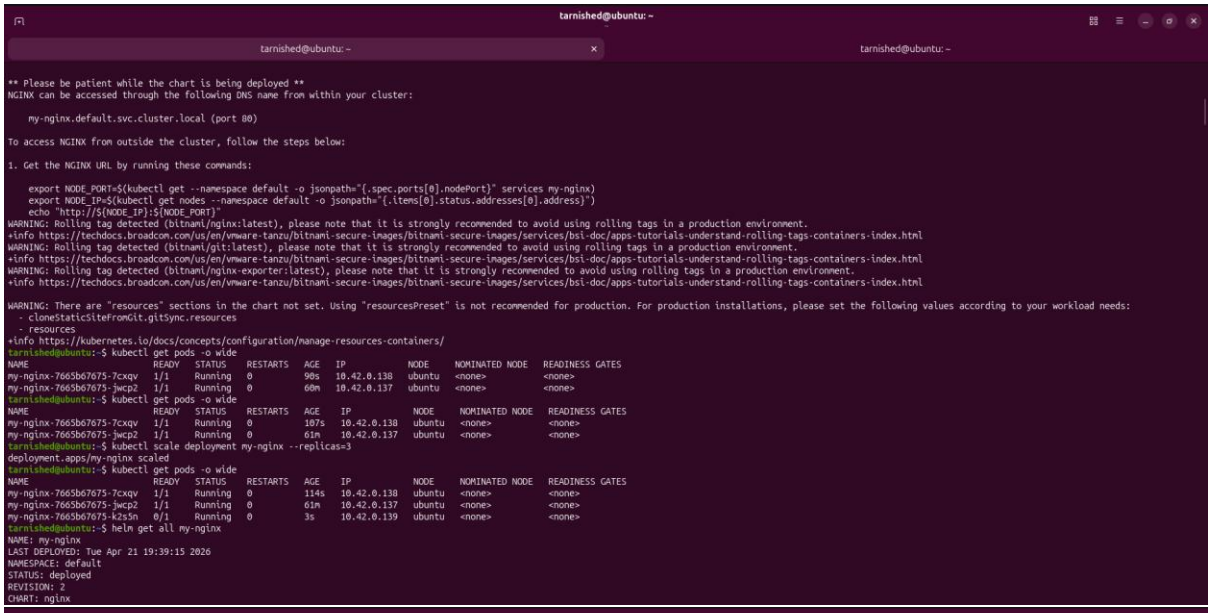
Course: Imtech

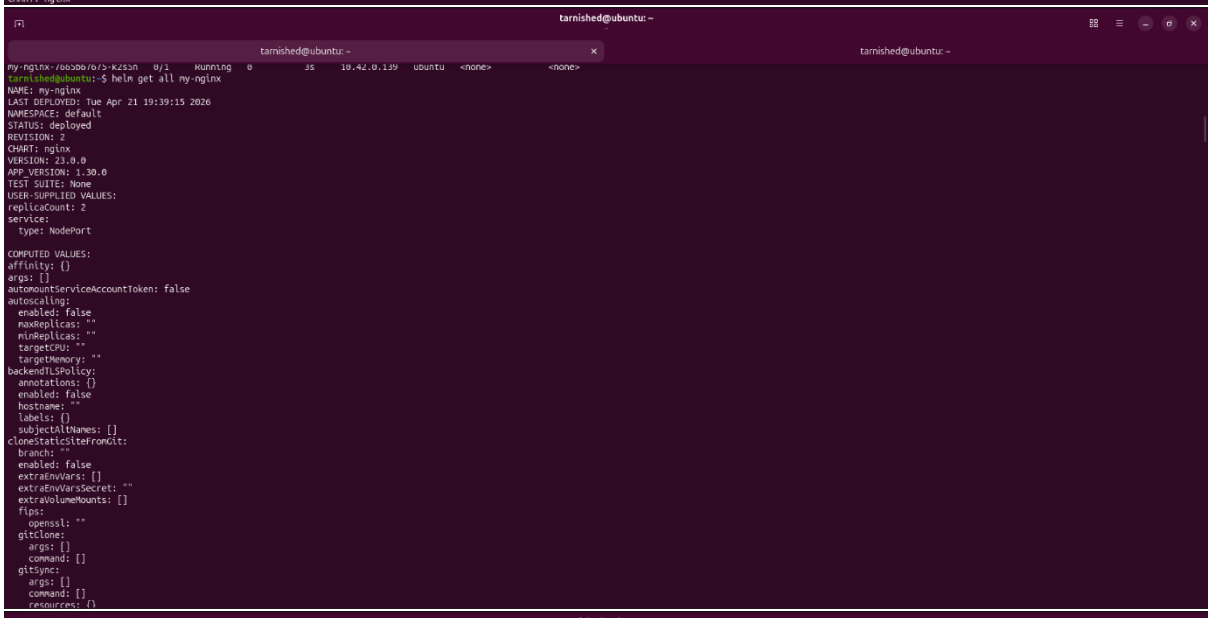
1.

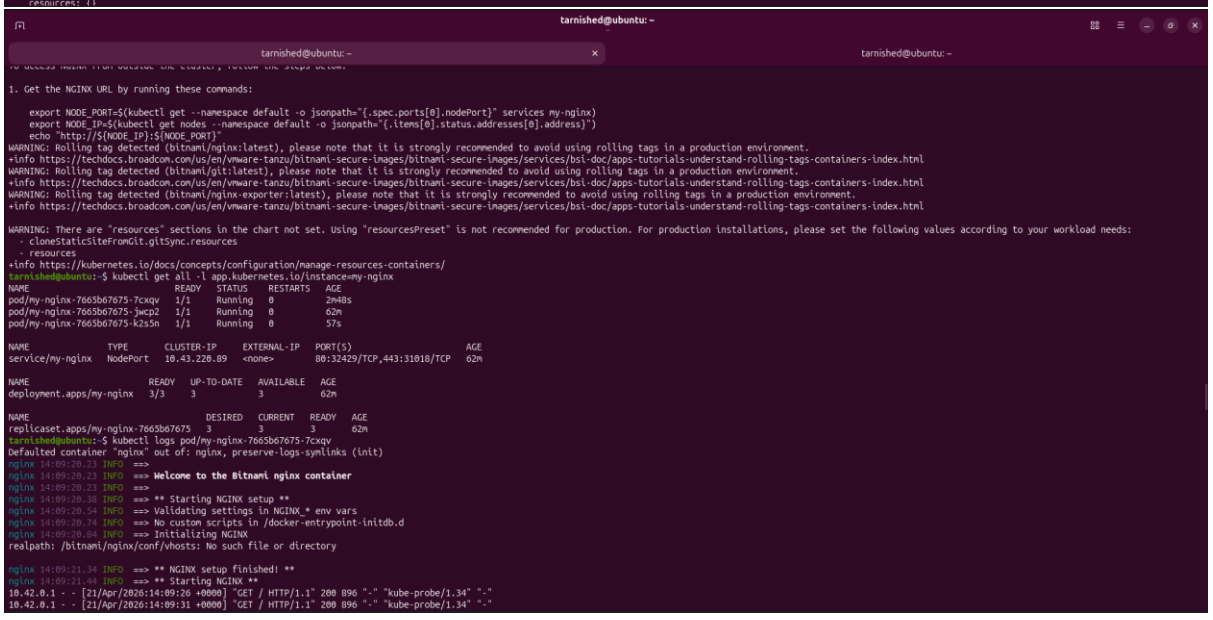
I already have helm and bitnami as we can see below and these are the series of commands I executed.











Below I sent ten requests in loop to check the requests.

```
tarnished@ubuntu:~$ for i in {1..10}; do curl http://10.0.2.15:32429; done
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
```

Below command kubectl logs -l app.kubernetes.io/instance=my-nginx will show all logs of the instance my-nginx and I have given individual too.

```
tarnished@ubuntu:~$ kubectl logs -l app.kubernetes.io/instance=my-nginx
Defaulted container "nginx" out of: nginx, preserve-logs-symlinks (init)
Defaulted container "nginx" out of: nginx, preserve-logs-symlinks (init)
Defaulted container "nginx" out of: nginx, preserve-logs-symlinks (init)
10.42.0.1 - - [21/Apr/2026:14:35:06 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:11 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:16 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:21 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:26 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:31 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:36 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:02 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:07 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:12 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:17 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:22 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:27 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:32 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:06 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:11 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:16 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:21 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:26 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:31 +0000] "GET / HTTP/1.1" 200 896 "-" "kube-probe/1.34" "-"
```

```
tarnished@ubuntu:~$ kubectl logs -f pod/my-nginx-7665b67675-k2s5n
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
^Ctarnished@ubuntu:~kubectl logs pod/my-nginx-7665b67675-jwcp2
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
tarnished@ubuntu:~$ kubectl logs pod/my-nginx-7665b67675-7cxqv
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
10.42.0.1 - - [21/Apr/2026:14:35:29 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.14.1" "-"
tarnished@ubuntu:~$ helm delete my-nginx
release "my-nginx" uninstalled
tarnished@ubuntu:~$ kubectl get pods -o wide
No resources found in default namespace.
tarnished@ubuntu:~$
```

2.

```
tarnished@ubuntu:~$ helm repo add openfaas https://openfaas.github.io/faas-netes/
"openfaas" has been added to your repositories
tarnished@ubuntu:~$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "openfaas" chart repository
...Successfully got an update from the "bitnami" chart repository
Update Complete. ⎈Happy Helming!⎈
tarnished@ubuntu:~$ kubectl create namespace openfaas
namespace/openfaas created
tarnished@ubuntu:~$ kubectl create namespace openfaas-fn
namespace/openfaas-fn created
tarnished@ubuntu:~$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   14d
kube-node-lease   Active   14d
kube-public       Active   14d
kube-system       Active   14d
openfaas          Active   63s
openfaas-fn       Active   37s
tarnished@ubuntu:~$ helm install openfaas openfaas/openfaas \
  --namespace openfaas \
  --set generateBasicAuth=true \
  --set faasnetes.readTimeout=30s \
  --set faasnetes.writeTimeout=30s \
  --set gateway.upstreamTimeout=30s \
  --set gateway.readTimeout=30s \
  --set gateway.writeTimeout=30s \
  --set queueWorker.ackWait=30s \
  --set openfaasPro=false
NAME: openfaas
LAST DEPLOYED: Tue Apr 21 20:36:01 2026
NAMESPACE: openfaas
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
To verify that openfaas has started, run:

  kubectl -n openfaas get deployments -l "release=openfaas, app=openfaas"

To retrieve the admin password, run:

  echo $(kubectl -n openfaas get secret basic-auth -o jsonpath="{.data.basic-auth-password}" | base64 --decode)
```

```
tarnished@ubuntu:~$ kubectl -n openfaas get deployments -l "release=openfaas, app=openfaas"
NAME           READY   UP-TO-DATE   AVAILABLE   AGE
alertmanager   1/1     1            1           76s
gateway        1/1     1            1           76s
nats           1/1     1            1           76s
prometheus     1/1     1            1           76s
queue-worker   1/1     1            1           76s
tarnished@ubuntu:~$ kubectl get pods -o wide -n openfaas
NAME                            READY   STATUS    RESTARTS        AGE     IP            NODE     NOMINATED NODE   READINESS GATES
alertmanager-fb97cfc46-h7krl    1/1     Running   0               2m31s   10.42.0.142   ubuntu   <none>           <none>
gateway-85d66c54b-6xlt2         2/2     Running   1 (2m18s ago)   2m31s   10.42.0.140   ubuntu   <none>           <none>
nats-5bf8cfb54-m29x9            1/1     Running   0               2m31s   10.42.0.143   ubuntu   <none>           <none>
prometheus-85cb68fd7b-hj9mm     1/1     Running   0               2m31s   10.42.0.141   ubuntu   <none>           <none>
queue-worker-66c8dd8dbb-c9g6x   1/1     Running   3 (116s ago)    2m31s   10.42.0.144   ubuntu   <none>           <none>
tarnished@ubuntu:~$ kubectl get deployment queue-worker -n openfaas -o yaml
```

```
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  revisionHistoryLimit: 10
```

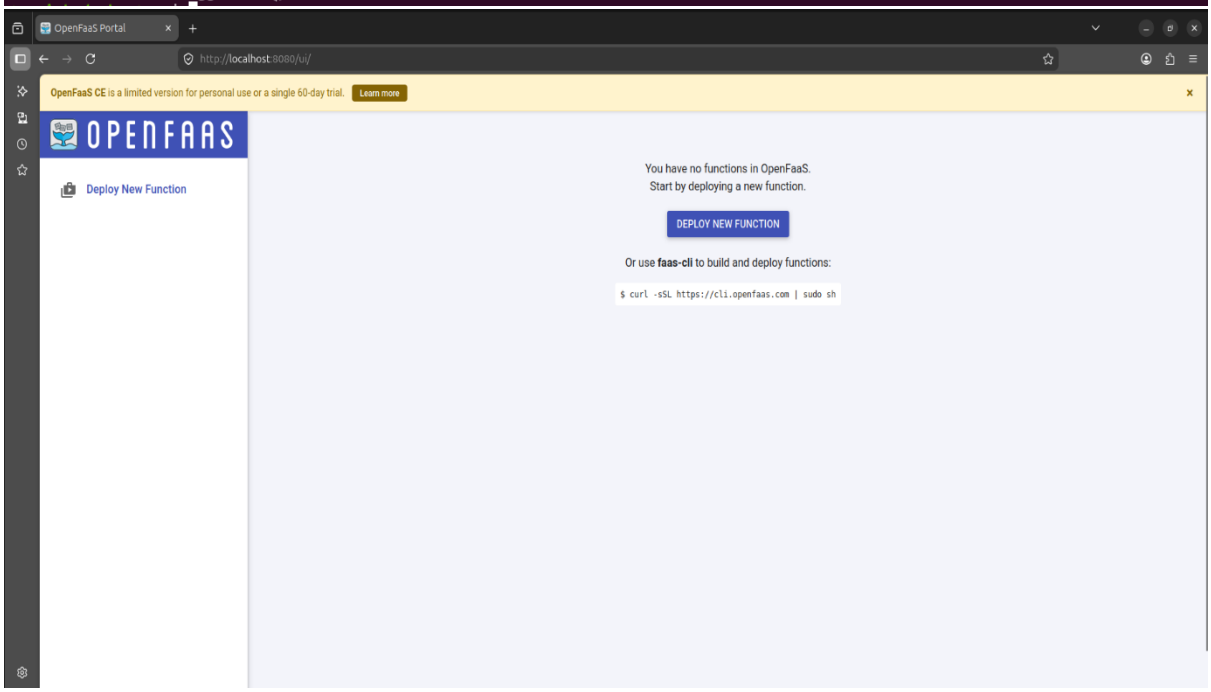
```
status:
  availableReplicas: 1
  conditions:
  - lastTransitionTime: "2026-04-21T15:06:02Z"
    lastUpdateTime: "2026-04-21T15:06:14Z"
    message: ReplicaSet "queue-worker-66c8dd8dbb" has successfully progressed.
    reason: NewReplicaSetAvailable
    status: "True"
    type: Progressing
  - lastTransitionTime: "2026-04-21T15:07:08Z"
    lastUpdateTime: "2026-04-21T15:07:08Z"
    message: Deployment has minimum availability.
    reason: MinimumReplicasAvailable
    status: "True"
    type: Available
  observedGeneration: 1
  readyReplicas: 1
  replicas: 1
  updatedReplicas: 1
tarnished@ubuntu:~$ kubectl get deployment queue-worker -n openfaas -o yaml | grep "replicas:"
  replicas: 1
  replicas: 1
```

```
tarnished@ubuntu:~$ kubectl get deployment alertmanager -n openfaas -o yaml
```

```
spec:
  progressDeadlineSeconds: 600
  replicas: 1
```

```
  readyReplicas: 1
  replicas: 1
  updatedReplicas: 1
tarnished@ubuntu:~$ kubectl get deployment alertmanager -n openfaas -o yaml | grep "replicas:"
  replicas: 1
  replicas: 1
tarnished@ubuntu:~$ kubectl get deployment gateway -n openfaas -o yaml
```

```
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  readyReplicas: 1
  replicas: 1
  updatedReplicas: 1
tarnished@ubuntu:~$ kubectl get deployment gateway -n openfaas -o yaml | grep "replicas:"
  replicas: 1
  replicas: 1
tarnished@ubuntu:~$ kubectl get deployment nats -n openfaas -o yaml
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  readyReplicas: 1
  replicas: 1
  updatedReplicas: 1
tarnished@ubuntu:~$ kubectl get deployment nats -n openfaas -o yaml | grep "replicas:"
  replicas: 1
  replicas: 1
tarnished@ubuntu:~$ kubectl get deployment prometheus -n openfaas -o yaml
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  readyReplicas: 1
  replicas: 1
  updatedReplicas: 1
tarnished@ubuntu:~$ kubectl get deployment prometheus -n openfaas -o yaml | grep "replicas:"
  replicas: 1
  replicas: 1
tarnished@ubuntu:~$ echo $(kubectl -n openfaas get secret basic-auth -o jsonpath="{.data.basic-auth-password}" | base64 --decode)
1e4HTc9Zvn0NsYIiJrf6YggC536BWQp3
```



```
tarnished@ubuntu:~$ kubectl port-forward -n openfaas svc/gateway 8080:8080 &
[1] 18546
tarnished@ubuntu:~$ Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
Handling connection for 8080
Handling connection for 8080
Handling connection for 8080
Handling connection for 8080
Handling connection for 8080
Handling connection for 8080
^C
tarnished@ubuntu:~$ curl -sSL https://cli.openfaas.com | sudo sh
[sudo: authenticate] Password:
Finding latest version from GitHub
0.18.8
Downloading package https://github.com/openfaas/faas-cli/releases/download/0.18.8/faas-cli as /tmp/faas-cli
Download complete.

Running with sufficient permissions to attempt to move faas-cli to /usr/local/bin
New version of faas-cli installed to /usr/local/bin
Creating alias 'faas' for 'faas-cli'.
 ___                   _____           ____
 / _ \ _ __   ___ _ __ |  ___|_ _  __ _/ ___|
| | | | '_ \ / _ \ '_ \| |_ / _` |/ _` \___ \
| |_| | |_) |  __/ | | |  _| (_| | (_| |___) |
 \___/| .__/ \___|_| |_|_|  \__,_|\__,_|____/
      |_|

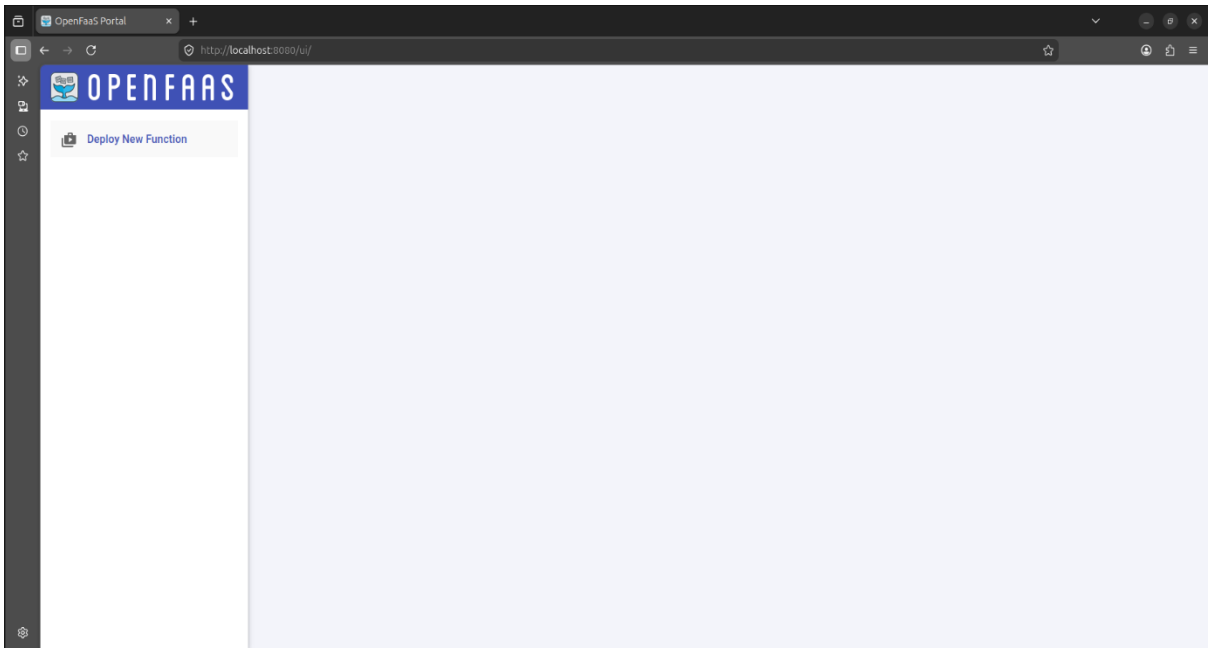
CLI:
 commit:  bf309592e77386f69cada2f4ceecac9f35f6d206
 version: 0.18.8
Handling connection for 8080
```

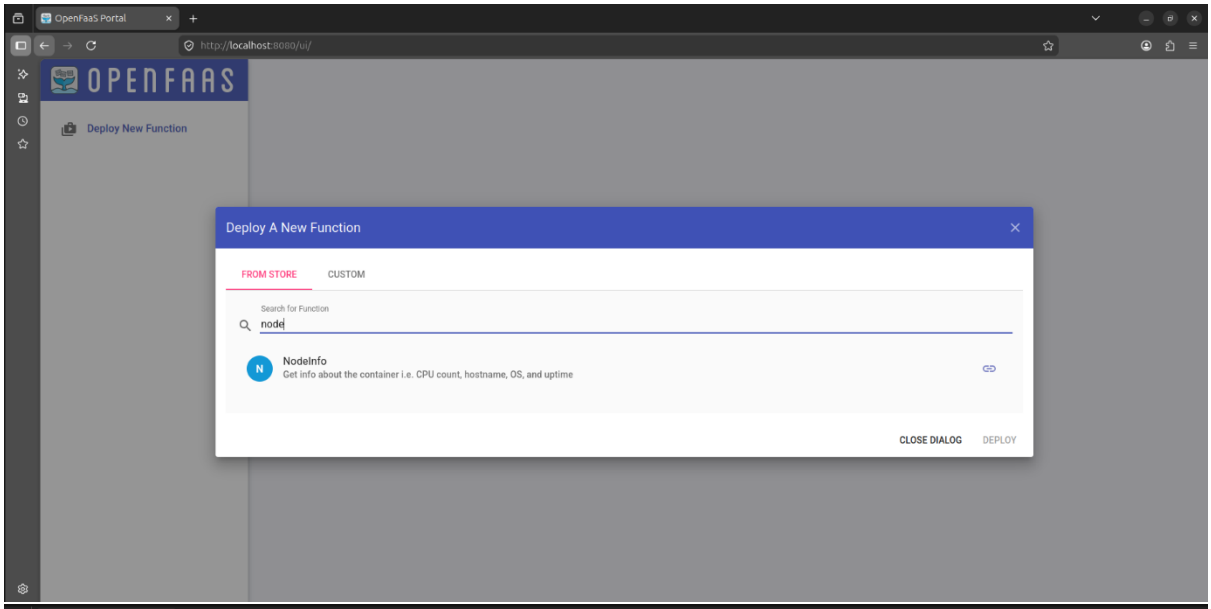
```
tarnished@ubuntu:~$ faas-cli version
 ___                   _____           ____
 / _ \ _ __   ___ _ __ |  ___|_ _  __ _/ ___|
| | | | '_ \ / _ \ '_ \| |_ / _` |/ _` \___ \
| |_| | |_) |  __/ | | |  _| (_| | (_| |___) |
 \___/| .__/ \___|_| |_|_|  \__,_|\__,_|____/
      |_|

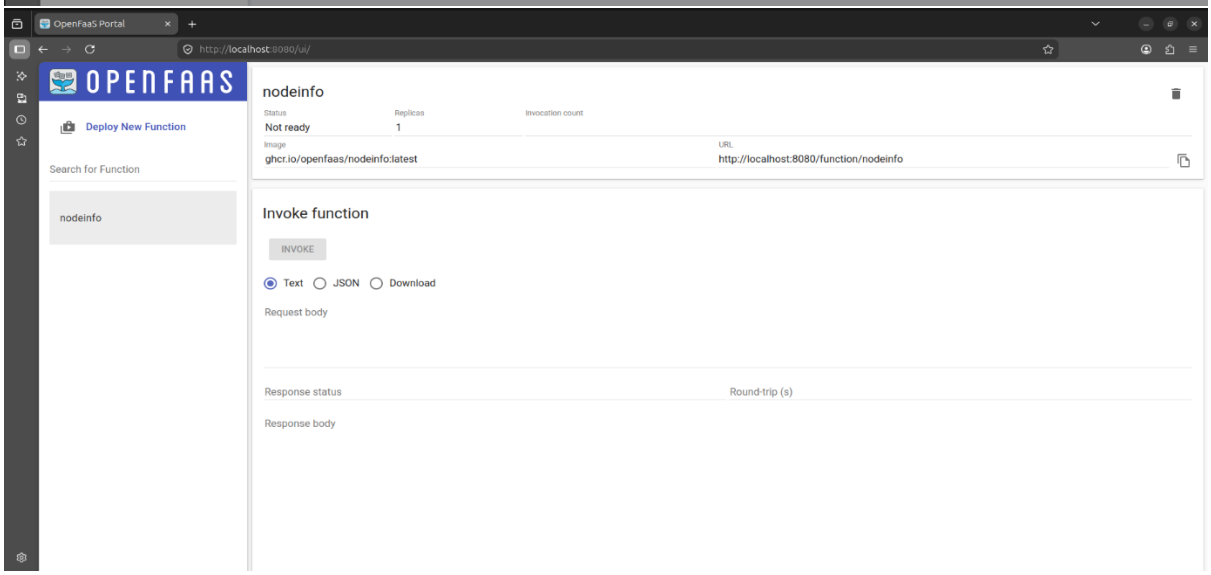
CLI:
 commit:  bf309592e77386f69cada2f4ceecac9f35f6d206
 version: 0.18.8
Handling connection for 8080
tarnished@ubuntu:~$ echo -n "1e4HTc9Zvn0NsYIiJrf6YggC536BWQp3" | faas-cli login --gateway http://127.0.0.1:8080 -u admin --password-stdin
Calling the OpenFaaS server to validate the credentials...
Handling connection for 8080
credentials saved for admin http://127.0.0.1:8080
tarnished@ubuntu:~$
```

3.

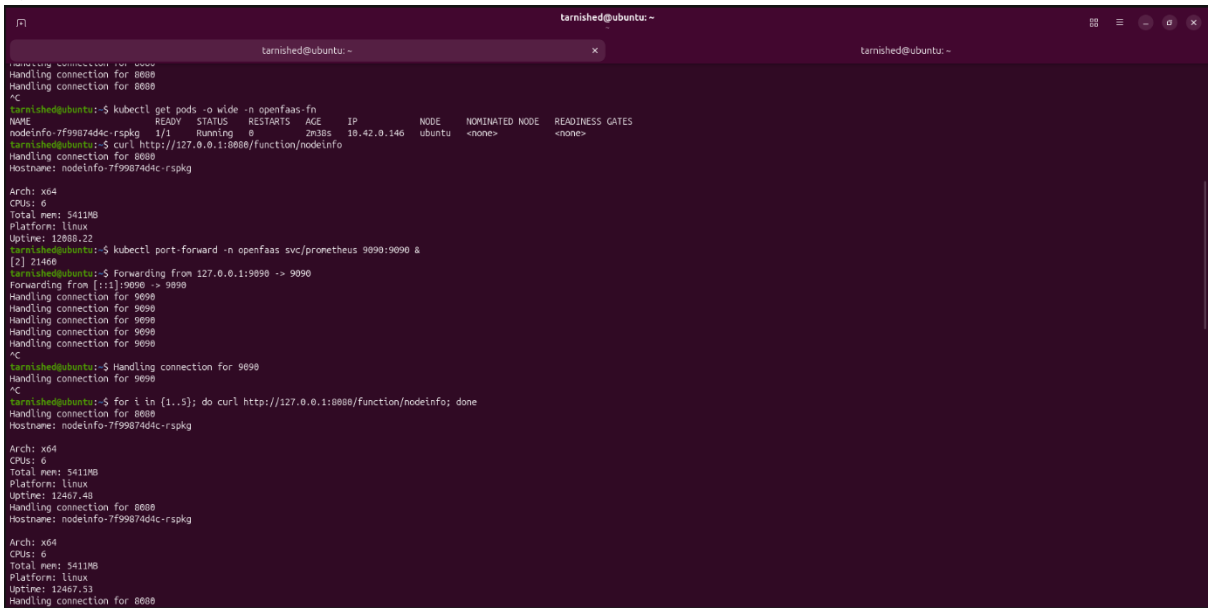
c)

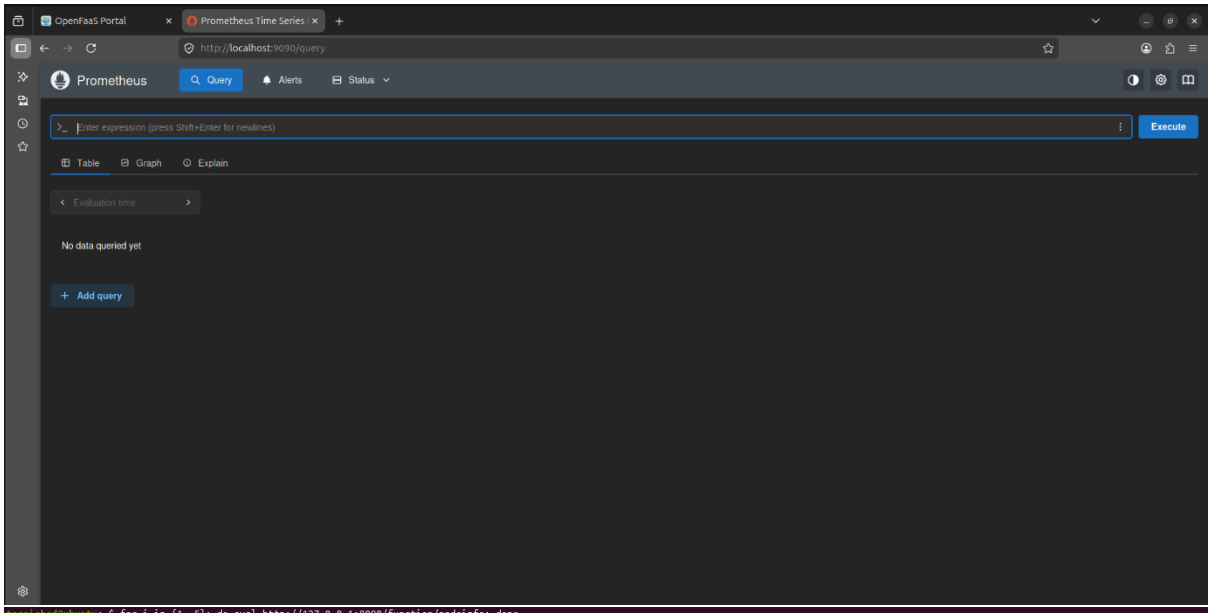



OpenFaaS Portal
http://localhost:8080/ui/
OPENFAAS
Deploy New Function
Deploy A New Function
FROM STORE
CUSTOM
Search for Function
node
NodeInfo
Get info about the container i.e. CPU count, hostname, OS, and uptime
CLOSE DIALOG
DEPLOY


OpenFaaS Portal
http://localhost:8080/ui/
OPENFAAS
Deploy New Function
Search for Function
nodeinfo
nodeinfo
Status
Not ready
Replicas
1
Invocation count
Image
ghcr.io/openfaas/nodeinfo:latest
URL
http://localhost:8080/function/nodeinfo
Invoke function
INVOKE
Text
JSON
Download
Request body
Response status
Round-trip (s)
Response body

e)


tarnished@ubuntu: ~
tarnished@ubuntu: ~
tarnished@ubuntu: ~
Handling connection for 8080
Handling connection for 8080
^C
tarnished@ubuntu:~$ kubectl get pods -o wide -n openfaas-fn
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
nodeinfo-7f99874d4c-rspkg 1/1 Running 0 2m38s 10.42.0.146 ubuntu <none> <none>
tarnished@ubuntu:~$ curl http://127.0.0.1:8080/function/nodeinfo
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg
Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12008.22
tarnished@ubuntu:~$ kubectl port-forward -n openfaas svc/prometheus 9090:9090 &
[2] 21460
tarnished@ubuntu:~$ Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
^C
tarnished@ubuntu:~$ Handling connection for 9090
Handling connection for 9090
^C
tarnished@ubuntu:~$ for i in {1..5}; do curl http://127.0.0.1:8080/function/nodeinfo; done
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg
Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12467.48
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg
Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12467.53
Handling connection for 8080




```
tarnished@ubuntu:~$ for i in {1..5}; do curl http://127.0.0.1:8080/function/nodeinfo; done
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg

Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12614.16
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg

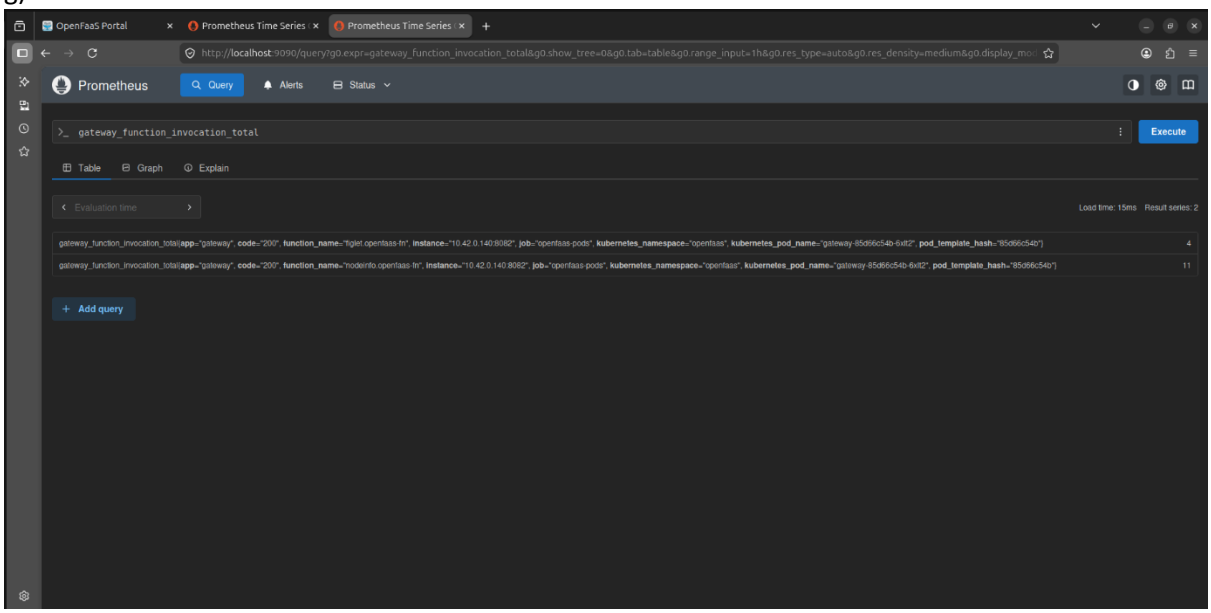
Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12614.2
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg

Arch: x64
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Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg

Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12614.29
Handling connection for 8080
Hostname: nodeinfo-7f99874d4c-rspkg

Arch: x64
CPUs: 6
Total mem: 5411MB
Platform: linux
Uptime: 12614.33
```

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GNU nano 8.4 handler.py

```
import cv2
import numpy as np
from ultralytics import YOLO
from flask import jsonify
import os
import uuid

model = YOLO("yolov5s.pt")

SAVE_DIR = "/home/app/received"
os.makedirs(SAVE_DIR, exist_ok=True)

def handle(req_data):
    try:
        if not req_data or len(req_data) == 0:
            return jsonify({"error": "Empty request"}), 400

        image = np.frombuffer(req_data, np.uint8)
        img = cv2.imdecode(image, cv2.IMREAD_COLOR)

        if img is None:
            return jsonify({"error": "Invalid image"}), 400

        results = model(img)

        detections = []

        for result in results:
            for box in result.boxes.data:
                x1, y1, x2, y2, conf, cls = box.tolist()

                start_point = (int(x1), int(y1))
                end_point = (int(x2), int(y2))
                color = (0, 255, 0)
                thickness = 2

                cv2.rectangle(img, start_point, end_point, color, thickness)

                label = model.names[int(cls)]

                cv2.putText(
```

GNU nano 8.4 handler.py

```
                cv2.rectangle(img, start_point, end_point, color, thickness)

                label = model.names[int(cls)]

                cv2.putText(
                    img,
                    f"{label} {conf:.2f}",
                    (int(x1), int(y1)-10),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    0.5,
                    color,
                    thickness
                )

                detections.append({
                    "x1": int(x1),
                    "y1": int(y1),
                    "x2": int(x2),
                    "y2": int(y2),
                    "confidence": float(conf),
                    "class": int(cls),
                    "label": label
                })

        filename = f"detected_{uuid.uuid4().hex}.jpg"
        filepath = os.path.join(SAVE_DIR, filename)

        success = cv2.imwrite(filepath, img)

        if not success:
            return jsonify({"error": "Failed to save image"}), 500

        return jsonify({
            "detections": detections,
            "saved_image_path": filepath
        })

    except Exception as e:
        return jsonify({"error": str(e)}), 500
```